

Dante Buhl

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RESEARCH INTERESTS

Fluid dynamics · rotating stratified turbulence · multiscale nonlinear dynamics · geophysical & astrophysical flows · high-performance scientific computing · PDE-based modeling · scientific machine learning

EDUCATION

Ph.D. in Applied Mathematics — University of California, Santa Cruz Sept 2024 – Present

Advisor: Prof. Pascale Garaud · Graduate GPA: 4.0/4.0 · Expected 2028

M.S. in Scientific Computing & Applied Mathematics — UC Santa Cruz Aug 2023 – Aug 2024

Thesis: “The Effect of Rotation on Stratified Turbulence” (Advisor: Prof. Pascale Garaud) · GPA: 4.0/4.0

B.A. in Mathematics, Highest Honors — UC Santa Cruz Sept 2020 – June 2023

Concentration: Computational Mathematics · GPA: 3.74/4.0

RESEARCH EXPERIENCE

Graduate Research Assistant — UC Santa Cruz, Applied Mathematics Sept 2023 – Present

Advisor: Prof. Pascale Garaud

- Leveraging multiscale models of nonrotating stratified turbulence to develop mixing models and measure turbulence.
- Investigating the effect of rotation on stratified turbulence in geophysical and astrophysical flows.
- Develop and run MPI-parallel Fortran simulations on high-performance computing clusters; analyze large simulation datasets to extract statistical scaling laws for turbulent transport.

Master’s Thesis — UC Santa Cruz, Applied Mathematics Fall 2023 – Summer 2024

“The Effect of Rotation on Stratified Turbulence” (Advisor: Prof. Pascale Garaud)

- Characterized how rotation modifies energy transport and mixing in stratified turbulent flows through direct numerical simulation.

Undergraduate Research — UC Santa Cruz, Mathematics 2023

- “The Pow-der of Ani-snow-tropy”: investigation of the Lorenz system, from *The Essence of Chaos* (E. N. Lorenz).

Research Assistant — Towson University Mathematics REU Summer 2023

- “Simulating Spherical Ciliated Organism Locomotion”: developed machine-learning methods to solve the Navier–Stokes equations in a Stokes-flow regime with spherical boundary conditions (Python).

Directed Reading Program — UC Santa Cruz, Mathematics Winter 2023

Mentor: Dr. John G. Pelias

- “Boxes and Balls: Quantifying Chaos”: numerical methods for computing the fractal dimension of chaotic attractors.

INDEPENDENT PROJECTS

Statistical Arbitrage via Ornstein–Uhlenbeck Mean Reversion Fall 2026

Independent study in stochastic methods for quantitative finance · Python

- Modeling spreads of cointegrated asset pairs as Ornstein–Uhlenbeck processes; deriving the exact discretization and maximum-likelihood estimators for the mean-reversion rate, long-run mean, and volatility.
- Assessing numerical and statistical accuracy: strong convergence of Euler–Maruyama against exact simulation, and small-sample bias in mean-reversion estimates.
- Selecting pairs with Engle–Granger cointegration tests and multiple-testing controls; estimating time-varying hedge ratios with a Kalman filter.
- Evaluating z-score trading rules strictly out of sample with transaction costs, including analysis of periods

where pair relationships break down.

PRESENTATIONS

EGU General Assembly 2025 — Stratified Turbulence in Geophysical & Astrophysical Flows 2025

Poster: “Rotating Stratified Turbulence.” Coauthors: P. Garaud, H. Wang.

APS Division of Fluid Dynamics 2024 — Geophysical Fluid Dynamics: Stratified Flows II 2024

Talk: “Rotating Stratified Turbulence.” Coauthors: P. Garaud, H. Wang.

TEACHING EXPERIENCE

Teaching Assistant — UC Santa Cruz, Applied Mathematics 2023 – Present

- AM 147 — Computational Methods and Applications (Winter 2026)
- AM 114 — Introduction to Dynamical Systems (Fall 2025)
- AM 112 — Introduction to Partial Differential Equations (Winter 2025)
- AM 11B — Mathematical Methods for Economists II (Spring 2024)
- Math 19B — Calculus for Science, Engineering & Mathematics II (Winter 2024)
- Math 19A — Calculus for Science, Engineering & Mathematics (Fall 2023)

Peer-Group Tutor — UC Santa Cruz, Learning Support Services 2022 – 2023

- Math 19A — Calculus (Fall 2022); Math 11A — Calculus with Applications (Winter–Spring 2023).

AWARDS & HONORS

- Next Generation Scholars in Applied Mathematics, UC Santa Cruz (2023)
- T&LC Graduate Pedagogy Fellowship, UC Santa Cruz
- B.A. with Highest Honors in Mathematics (2023)
- Dean’s Honors, UC Santa Cruz (multiple quarters)

GRADUATE COURSEWORK (SELECTED)

Advanced Methods of Applied Mathematics (A+) · Applied & Advanced PDEs (A+) · Numerical Linear Algebra (A+) · Numerical Methods for Differential Equations (A+) · High Performance Computing (A+) · Magnetohydrodynamics (A+) · Advanced Fluid Dynamics (A) · Computational Fluid Dynamics (A) · Stochastic Differential Equations (A) · Scientific Machine Learning (A) · Applied Dynamical Systems (A)

TECHNICAL SKILLS

Programming: Python, Fortran, MATLAB, Java, JavaScript, R

Methods: Scientific machine learning, numerical methods, numerical linear algebra, stochastic methods, PDE modeling, dynamical systems

Computing: MPI parallel programming, high-performance & cluster computing (Linux), Git, L^AT_EX

Languages: English (native), Spanish (intermediate)

SERVICE & ACTIVITIES

- Secretary & Member, Slugs United by Mathematics (Math Club), UC Santa Cruz (2022–2024).

REFERENCES

Prof. Pascale Garaud

Chair & Professor of Applied Mathematics, UC Santa Cruz

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Prof. Hongyun Wang

Professor of Applied Mathematics, UC Santa Cruz

UC Santa Cruz, 606 Engineering Loop, Santa Cruz, CA 95064

Prof. Chris Edwards

Professor of Ocean & Earth Sciences, UC Santa Cruz

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